QUESTION 1

Which of the following is not a function of timers in microcontrollers?

Answer is Correct **Score: 1 / 1**

Options

A. Generating arbitrary delays

B. Measuring signal frequency

Correct

C. Storing program instructions permanently

D. Timekeeping

FEEDBACK

Solution: Timers can generate delays, measure frequencies, and keep time, but storing program instructions is the role of memory, not timers.

Link: https://youtu.be/rox5q8nYz4E?feature=shared&t=29

QUESTION 2

Why are timers in STM32 microcontrollers useful for power management?

Not Attempted **Score: 0 / 1**

Options

A. They consume no power during operation

Expected

B. They operate independently of the CPU

C. They store more data than regular memory

D. They require no clock source

FEEDBACK

Solution: Timers run independently of the CPU, allowing the CPU to enter deep sleep while the timer continues counting and can wake the CPU when needed.

Link: https://youtu.be/rox5q8nYz4E?feature=shared&t=157

QUESTION 3

What is the role of the prescaler in a timer?

Not Attempted **Score: 0 / 1**

Options

A. To increase the counting speed

Expected

B. To divide the input clock frequency to a lower rate

C. To store timer values for later use

D. To reset the timer automatically

FEEDBACK

Solution: The prescaler reduces the input clock frequency so that the timer can operate over a longer counting period.

Link: https://youtu.be/rox5q8nYz4E?feature=shared&t=243

QUESTION 4

In the context of watchdog timers, what does “patting the watchdog” mean?

Not Attempted **Score: 0 / 1**

Options

A. Stopping the timer from starting

Expected

B. Resetting the watchdog timer count before it reaches zero

C. Increasing the watchdog timeout automatically

D. Turning off the watchdog permanently

FEEDBACK

Solution: Patting the watchdog means resetting it to its initial value at regular intervals so it doesn’t trigger a system reset.

Link: https://youtu.be/rox5q8nYz4E?feature=shared&t=724

QUESTION 5

Which of the following is a drawback of using a watchdog timer?

Not Attempted **Score: 0 / 1**

Options

A. It reduces system reliability

Expected

B. It can cause false resets if the timeout is too short

C. It prevents automatic recovery from errors

D. It increases debugging visibility

FEEDBACK

Solution: If the timeout is set too low, the watchdog may reset the system even when the CPU is still functioning but busy, leading to false resets.

Link: https://youtu.be/rox5q8nYz4E?feature=shared&t=867